Course structure of B.Tech (Civil Engineering) for batch 2024-28

Semester 5	OE		Open Elective 2 **((NPTEL/SWAYAM/MOOC))**	3	0	0	3	3
	HSC		Engineering Economics	3	0	0	3	3
	PE		Program Elective 1	3	0	0	3	3
	PC		Hydrology and Water Resources	3	0	2	5	4
	PC		Foundation and Geotechnical Applications	3	0	2	5	4
	PC		Design of RCC Structures	3	0	0	3	3
	PC		Computer Aided Design Drawing Lab - I	0	0	2	2	1
				18	**0**	**6**	**24**	**21**
Semester 6	OE		Open Elective 3 (From Other Department of FoET)	3	0	0	3	3
	PE		Program Elective 2	3	0	0	3	3
	PE		Program Elective 3	3	0	0	3	3
	PC		Design of Steel Structures	3	0	0	3	3
	PC		Computer Aided Design Drawing Lab - II	0	0	2	2	1
	PC		Environmental Engineering	3	0	2	5	4
	PC		Highway and Traffic Engineering	3	0	2	5	4
				18	**0**	**6**	**24**	**21**

Course structure of B.Tech (Civil Engineering) for batch 2024-28

Semester 7	Pro		Summer Internship	0	0	0	0	2
	OE		Open Elective 4 (From Other Department of FoET)	3	0	0	3	3
	PE		Program Elective 4	3	0	0	3	3
	PE		Program Elective 5	3	0	0	3	3
	PC		Estimation Costing Contracts and Valuations	3	1	0	4	4

Teaching Scheme					Hydrology and Water Resources Engineering					
					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the basic component of hydrology
2. To understand the fundamentals of Ground water hydrology
3. To understand the basic concept of reservoir operation and irrigation techniques
4. To understand the case application of modeling techniques in water resources engineering

UNIT I : INTRODUCTION TO COMPONENT OF HYDROLOGY **14 Hrs.**

Introduction - Hydrologic Cycle, Precipitation, Types - Measurement – Rain Hyetograph and Mass Curve - DAD – IDF curve - Run Off - Evaporation Losses, Infiltration: Process-Capacity-Rate-Capacity Curve-Indices -Computation of Runoff- Unit hydrograph, Flood hydrograph, S-Curve Hydrograph.

UNIT II : GROUND WATER HYDROLOGY **10 Hrs.**

Definition – Occurrence - Zone and Movement of GW - GW Yield - Aquifers and their Types - Thiem's and Dupuit's formulas - Efficiency of Well - Infiltration Galleries and Well - Introduction to Open and Tube Well.

UNIT III : RIVER TRAINING AND RESERVOIR PLANNING **12 Hrs.**

Introduction - Different type of river - classification of Indian river - mender parameters and their relationships – objective of river training Classification of river training work- Reservoir – River Capacity and Diversion Channel - Reservoir Planning, Definition and Types - Capacity-Elevation and Area-Elevation Curve - Storage Zones - Catchment and Reservoir Yield, Reservoir Capacity and Methods of Finding – Hydrologic Reservoir Routing and Methods - Reservoir Sedimentation, Reservoir Losses.

UNIT IV: MODELING TECHNIQUES IN WATER RESOURCES ENGINEERING **06Hrs.**

Introduction of model - Types of models, Introduction of Hydrologic and Hydrodynamic model - Different Case applications of modelling in water resources engineering. Drone application in water resource engineering.

TOTAL HOURS: 42 Hrs.

COURSE OUTCOMES:

CO-1: Define the basic components of hydrology cycle.

CO-2: Define the basics of Ground water hydrology.

CO-3: Evaluate components of hydrologic cycle.

CO-4: Classify the various river and reservoir systems for water resources management.

CO-5: Analyze different surface and subsurface components of hydrology.

CO-6: Elaborate the utility of various models in water resources engineering.

TEXT-BOOK AND REFERENCE BOOKS:

1. S.K. Garge, **"Hydrology and Water Resources Engineering"**, Khanna publishers, New Delhi.
2. H.M. Raghunath, **"Ground Water"**, Wiley publishers, New Delhi.
3. K. R. Arora, **"Irrigation, Water Power and Water Resources Engineering"**, Standard publishers, Delhi.
4. P. J. Rami Reddy, **"Hydrology"**, Laxmi publishers, New Delhi.

Teaching Scheme					Hydrology and Water Resources Engineering laboratory					
					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To understand the basic component of hydrology
2. To understand the fundamentals of Ground water hydrology
3. To understand the basic concept of reservoir operation and irrigation techniques
4. To understand the case application of modeling techniques in water resources engineering

LIST OF EXPERIMENTS

1 To measure the rainfall using the Tipping bucket rain-gauge.
2 To calculate an average rainfall using graphical (Thiessen Polygon) Method.
3 To estimate the rate of evaporation using Pan Evaporometer.
4 To estimate the rate of infiltration using single and double ring infiltrometer. .
5 To measure the velocity using the current meter.
6 To calculate the area of irregular boundary using a planimeter- A case of reservoir.
7 To study the hydraulic jump.
8 To study the various software for drainage delineation.
9 To study the various software for watershed delineation.
10 To study drone (UAV) for DEM/DTM generation – A case of 1D and 2D hydraulic modeling

COURSE OUTCOMES:

CO-1: Understand the basic components of hydrology cycle.
CO-2: Calculate the various hydrologic component through experiment.
CO-3: Evaluate area-volume using planimeter.
CO-4: Classify the river flow using hydraulic flume.
CO-5: create the watershed and drainage using a software techniques.
CO-6: Create a DEM/DSM using Drone (UAV) techniques for hydraulic modeling.

TEXT-BOOK AND REFERENCE BOOKS:

1. S.K. Garge, **"Hydrology and Water Resources Engineering"**, Khanna publishers, New Delhi.
2. H.M. Raghunath, **"Ground Water"**, Wiley publishers, New Delhi.
3. K. R. Arora, **"Irrigation, Water Power and Water Resources Engineering"**, Standard publishers, Delhi.
4. **"2D Modeling User's Manual",** HEC-RAS River Analysis System, US Army Corps of Engineers.

					Foundations and Geotechnical Applications					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	4	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To emphasize the importance of geotechnical investigation and subsoil exploration techniques.
2. To explain the importance of earth pressure and slope stability in geotechnical structures.
3. To compute bearing capacity for various foundation systems, including settlement considerations.
4. To explain the suitability of appropriate foundation systems for various site conditions and their analysis.

UNIT I: SUBSOIL EXPLORATION AND GEOTECHNICAL INVESTIGATION **10 Hrs.**

Objective, Subsurface exploration planning, Exploration methods, Soil sampling and samplers, In-situ tests, Geophysical exploration, Geotechnical investigation report, Problems and solutions.

UNIT II: EARTH PRESSURE THEORIES AND STABILITY OF SLOPES **12 Hrs.**

Introduction, Types of earth pressures, Rankine's theory, Surcharge, Inclined backfill, Soil stratification, Graphical methods, Application problems, and solutions, Types of slopes, Causes of failure, Stability analysis of infinite and finite slopes, Types of failures, slip circle method, friction circle method, Taylor's stability number and stability of curves.

UNIT III: BEARING CAPACITY AND FOUNDATION SETTLEMENT **10 Hrs.**

Definitions, Types of failures, Terzaghi's analysis, Skempton's formula, IS formula, Effect of water table on bearing capacity, Shape of foundation, Inclination of load and eccentricity of load on bearing capacity, Allowable bearing pressure, Plate load test, Penetration tests, Settlement of foundation: Settlement analysis, Types of foundation settlements, Components of settlements and their estimation, Allowable settlement values, Effects, causes and remedial measures of total and differential settlements.

UNIT IV: SHALLOW AND DEEP FOUNDATIONS **10 Hrs.**

Shallow foundations, Stress distribution (theory only): Rectangular combined footing, Trapezoidal combined footing, Raft footing, IS code of practice, Deep foundations, Functions of piles and their classification, Relative merits, Load carrying capacity of piles, Static and dynamic formula, Pile load test, Penetration tests, Pile spacing and group action, Design of pile group, Settlement of pile group. Negative skin friction, Under reamed pile, Load carrying capacity of pile group.

TOTAL HOURS: 42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO-1: Develop a subsurface investigation program.

CO-2: Calculate the earth pressure for different field conditions and soils.

CO-3: Analyze the stability of soil slopes.

CO-4: Compute the bearing capacity of foundations and foundation settlements.

CO-5: Analyse shallow foundations.

CO-6: Analyze deep foundations.

TEXT-BOOK AND REFERENCE BOOKS:

1. B.C.Punmia, Ashok Kumar Jain, Arun Jain, "**Soil Mechanics and Foundations**", Laxmi Publications.
2. K.R. Arora, "**Soil Mechanics and Foundation Engineering**", Standard Publishers.
3. Shashi Gulathi, Manoj Dutta, "**Geotechnical Engineering**", McGraw Hill Education.
4. Bowles J.E., "**Foundation Analysis and Design**", McGraw-Hill Education.
5. Gopal Ranjan, Rao A.S.R., "**Basic and Applied Soil Mechanics**", New Age International Pvt. Ltd.

					Foundations and Geotechnical Application Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	25	25	50

COURSE OBJECTIVES

1. To determine engineering properties of soils in the context of foundation design
2. To understand the soil exploration techniques
3. To determine in-situ parameters of soils by field test

LIST OF EXPERIMENTS

1. Determination of free swell index and swelling pressure of soil
2. Determination of CBR value of soils
3. Determination of shear strength parameters using triaxial compression test (undrained)
4. Determination of shear strength of soft soils using vane shear test
5. Determination of bearing capacity of foundation using plate load test
6. Demonstration of soil exploration and SPT, CPT test
7. Electrical resistivity of soils
8. Dynamic cone penetration test
9. Analysis of slope stability and bearing capacity of foundations
10. Laboratory based projects

COURSE OUTCOMES

CO-1: Determine free swell index and swelling pressure of soil
CO-2: Determine CBR value of soils
CO-3: Evaluate shear strength of soil.
CO-4: Interpret in-situ parameters conducting SPT,DCPT
CO-5: Determine bearing capacity parameters using plate load test
CO-6: Interpret field properties using geophysical test

TEXT-BOOK AND REFERENCE BOOKS:

1. Laboratory Manual
2. B.C.Punmia, Ashok Kumar Jain, Arun Jain, "**Soil Mechanics and Foundations**", Laxmi Publication
3. K.R. Arora, "**Soil Mechanics and Foundation Engineering**", Standard Publisher Dist.
4. Parbin Singh, "**Engineering and General Geology**", S.K. Kataria & Sons
5. Shashi Gulathi, Manoj Dutta, "**Geotechnical Engineering**", Tata Mcgraw Hill Publisher

					Design of RCC Structures					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES:

1. Understand the basics of structural design for reinforced concrete structures, including strength, stability, and serviceability, stress-strain curves, stress block.
2. Learn Limit State design methods and their advantages over other design philosophies like Ultimate Load and Working Stress methods. Beam classification based on reinforcement configurations.
3. Design reinforced concrete structural members, such as beams, slabs, staircases, columns, and footings, for various loading conditions according to Indian Standard codes.

UNIT I : INTRODUCTION **10 Hrs.**

Introduction - Introduction and scope - Design: strength – stability – serviceability - Design methods - Ultimate load method - working stress method - Limit state method - Standard Loadings - Partial safety factors for materials and loads – Grades of concrete - characteristic strength of concrete - yield strength of mild and Tor steels - Analysis of RC Beams (flexural members) - Stress-Strain curve for concrete - stress block for RC beam section - Classification of Beams based on reinforcement and depth of neutral axis - Moment resistance of RC-rectangular

UNIT 2 DESIGN OF RCC BEAMS **10 Hrs.**

Design of RC-beams - Rectangular - T and L beams for under- reinforced and balanced sections - & singly and doubly reinforced - Design of shear reinforcement - check for development length and deflection

UNIT 3 DESIGN OF RCC SLABS AND STAIRCASE **10 Hrs.**

Design of slabs - Design of one way, two-way and cantilever slabs for flexure and check for shear deflections Design Staircase - Types of Staircase – components – landing – going – tread – rise – loading – geometry - load calculations - Design of Dog legged staircase

UNIT VI: DESIGN OF COMPRESSION MEMBERS AND FOOTING **12 Hrs.**

Design of Compression Members - Effective lengths, slenderness ratio, Short and long columns definitions - Design of short columns subjected to axial load - uni-axial bending and biaxial bending moments Design of Footing - SBC of soil -design of flat and sloped square - rectangular and circular footings for RC columns subjected to axial load - uniaxial and biaxial bending moments - Check for shear

MAX. 42 Hrs.

COURSE OUTCOMES:

CO-1: Recall the fundamental principles of RCC structure design.
CO-2: Design RCC beams, including Rectangular, T, and L beams.
CO-3: Proficiency in designing RCC slabs for flexure and shear deflections.
CO-4: Design of staircases, considering components, geometry, and loading requirements.
CO-5: Competence in designing compression members.
CO-6: Demonstrate the ability to design various types of isolated footings.

TEXT-BOOK AND REFERENCES BOOKS:

1. P.C.Vergese, "**Limit State Design of Reinforced Concrete**", PHI Publications, New Delhi
2. N Krishna Raju, "**Design of Reinforced Concrete Structures**", CBS Publishers New Delhi
3. Ramamrutham, "**Design of Reinforced Concrete Structure"**, Dhanpat & Roy Publishers, New Delhi.
4. Shah H.J, "**Reinforced Concrete Volume-I**", Charotar Publication House Pvt .Ltd

5. V L Shah, S R Karvem, "**Limit State Theory & Design of Reinforced Concrete**",(I.S. 456 - 2000) Standard Publishers Distributors
6. **"IS: 456-2000, Indian Standard Plain and Reinforced Concrete Code of Practice**", Bureau of Indian Standards, New Delhi
7. "**SP:16(6)-1980, Design Aids for Reinforced Concrete to IS: 456-1978**", Bureau of Indian Standards, New Delhi

					Computer Aided Design Drawing Laboratory - I					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	00	00	00	50	50	100

COURSE OBJECTIVES:

1. To familiarize students with drafting tools for creating plans, elevations, and sections of residential buildings
2. To introduce students with Finite Element Method (FEM) software for analysis and design RCC structures.
3. To gain familiarity with bar bending schedules, layout and detailing of structural drawings

UNIT I : DRAWING OF RESIDENTIAL BUILDING **08 HRS**

Familiarising basic commands of drafting software To draw plan, elevation of a residential buildings using drafting software. To draw sectional elevation of the building specified in the above item using drafting software.

UNIT II : INTRODUCTION TO FEM SOFTWARE **06 HRS**

Introduction to FEM Software Pre-processing- developing models of simple structural member like simple and continuous beams, 2D trusses and 2D frames-Defining material properties-Application of boundary conditions like roller, hinged, fixed, partially restrained supports -Application of loads like point load, distributed load and concentrated moments, handling editor files. Processing running the simulations, and handling the errors and warnings Post-Processing reading the results, critical internal forces, plotting shear force diagrams, bending moment diagrams, axial force diagrams and tornal moments diagrams.

UNIT III : DESIGN OF RESIDENTIAL BUILDINGS USING FEM SOFTWARE **06 HRS**

Analysis and Design of Residential Buildings Structural Structures as Per IS 456, IS 1893
Developing/Importing model of G+2 building. Defining material properties-Application of boundary conditions, Partially and fully releasing the moments -Application of loads as Per IS 456, IS 1893, defining load combinations. Defining design parameters and assigning design commands. Preforming the analysis, reading the results

UNIT VI : DRAWING OF REINFORCED CONCRETE STRUCTURES **10 HRS**

Prepared typical layout drawings for various structural member like footing, column, beams, slabs with appropriate notations system. To draw longitudinal and cross sections of three span continuous beam showing the reinforcement details. To draw plan and sectional elevation of simply supported one way and two-way slabs showing the reinforcement details. To draw plan and sectional elevations of rectangular column with rectangular footings. To draw plan of the staircase room and sectional elevations of flights showing reinforcement details

MAX. 32 HRS

COURSE OUTCOMES

CO-1: Demonstrate proficiency in using drafting software tools to create building drawing.
CO-2: Create finite element models of simple structural members such as beams, trusses, and frames using FEM software.
CO-3: Demonstrate proficiency in using FEM software as a tool for structural analysis and design.
CO-4: Apply relevant design codes and standards (IS 456, IS 1893) in the analysis and design process
CO-5: Utilize analysis results to assess structural performance and make informed design decisions
CO-6: Demonstrate professionalism in technical drawing practices, including adherence to standards and conventions

TEXT-BOOK AND REFERENCES BOOKS:

1. **"2D CAD Drawing User's Manual",** Autodesk, US.